

The use of Containers in OpenGL, ML and HPC for Teaching and Research Support

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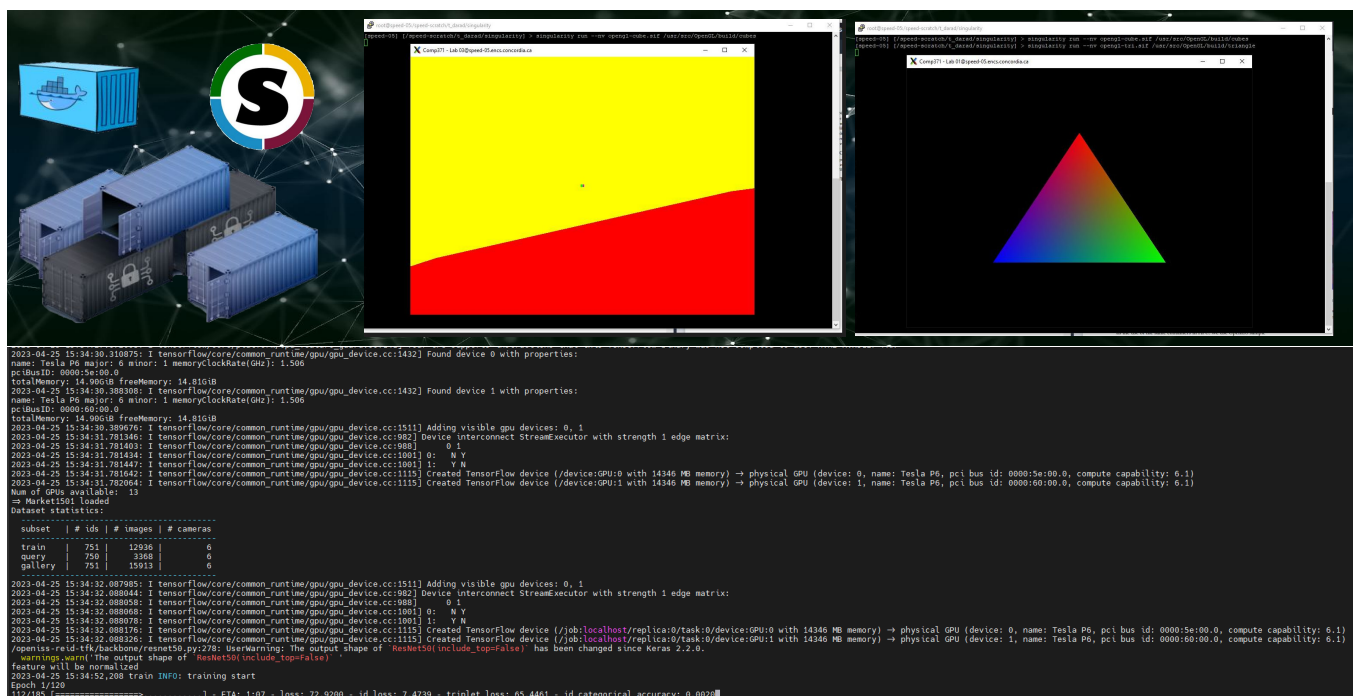


Figure 1: Sample teaser image of OpenGL in containers.

ABSTRACT

We share our experience of using containers (Docker and Singularity) in computer graphics teaching and research support, as well as the use of the same containers in HPC. We use OpenISS sample containers for this purpose, containers that are open-source and publicly available.

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CCS CONCEPTS

• **Software and its engineering** → **Middleware**; • **Human-centered computing** → **Visualization systems and tools**; • **Computing methodologies** → **Rendering**.

KEYWORDS

containers, OpenGL, graphics programming, deep learning, HPC

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1 OVERVIEW

We prepared a number of OpenISS-based containers in order to test the feasibility of using them for OpenGL projects, CUDA, machine/deep-learning projects that make use of GPUs, Jupyter notebooks, Conda, and other common libraries. The purpose of the containers research and implementation was to have a common, consistent environment that both instructors and students could reliably reproduce regardless of their preferred OS. The container middleware was made available on lab desktops and the students were given access to containers. This is a collaboration between the IT team as well as faculty members and students. Docker was

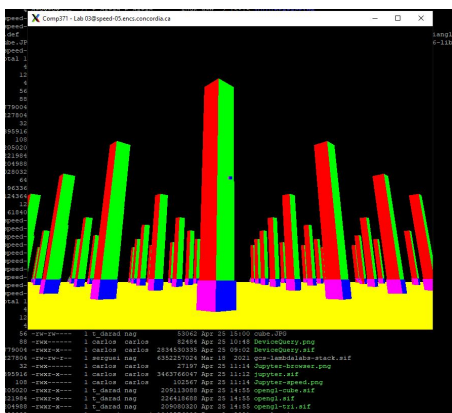


Figure 2: Interactive OpenGL Cube in a Container on Speed

the first type of container that was offered as a proof of concept using OpenISS images. The container implemented during this proof of concept went on to support an actual course of the Winter semester – COMP371 – Computer Graphics. The other container type is Singularity, which is more HPC-oriented. We were able to convert our Docker images into Singularity images and offer the same exact containers to the HPC community via our HPC cluster Speed. Fidelity of the containers for both batch and interactive process was demonstrated and made available. The interactive mode is important for use cases that require the rendering of high-fidelity OpenGL projects combined with deep learning components in their pipeline. As a sample we incorporated an OpenISS wrapper for poin2mesh [Hanocka et al. 2020] by Hanocka et al. (from the past SIGGRAPH) among others. We also integrated the REID example by Eric Lai [Lai 2019]. The integration of the containers into the teaching had its challenges. For example, the specialized teaching lab desktops were Windows-only without dual-boot Linux support. Therefore, running Docker Desktop on the Windows environment allowed for the delivery of the prepared course content consistently. We have found a proper way to automate some of the tasks and allow the use of the GPUs on Windows from within Linux containers and a proper graphics display between the two systems. We were likewise able to test CUDA devices, and NVIDIA samples, and OpenGL samples and offer labs as such to the students. The disadvantage of the use of this technology in a course content like that was that the students had to learn a new technology that does not necessarily directly relate to the computer graphics content

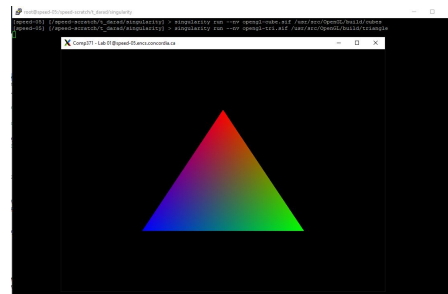


Figure 3: Interactive OpenGL Cube in a Container on Speed

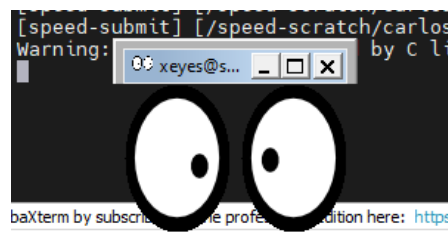


Figure 4: Xeyes on Speed

they are learning. However, in the longer term, when the future is moving towards containers for continuous integration (CI) and reproducibility, this approach is promising to be more sustainable. The project has been extended to run in HPC environments where for example the same OpenISS REID application can run in a job format in an HPC, using the exact same interactive container. A sample output from a training session can be seen in Figure 1.

2 CONCLUSION

We provided a solution to the issue of reproducibility, Linux environment, GPU and CUDA access, graphics rendering, for teaching and research in desktop and HPC environments using the same container images and their open specifications. This is especially useful if the labs and assignments are given remotely or all team members in a project team need the same consistent integrated environment. The actual project is hosted here: <https://github.com/OpenISS/openiss-dockerfiles>, and its HPC fork is here: <https://github.com/NAG-DevOps/openiss-dockerfiles>. And the related container specifications for COMP371 are here: https://github.com/tiperiu/COMP371_dockerimage.

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